

Climate Risk & Vulnerability Assessment Report

Asset Name:	Phoenix Data Center II	Assessment Date:	2026-06-11 10:23
Asset Type:	Data Center	Latitude / Longitude:	33.495004, -111.987013
Address:	4100, North 44th Street, Camelback East, Phoenix, Maricopa County, Arizona, 85018, United States	Climate Scenario:	SSP585

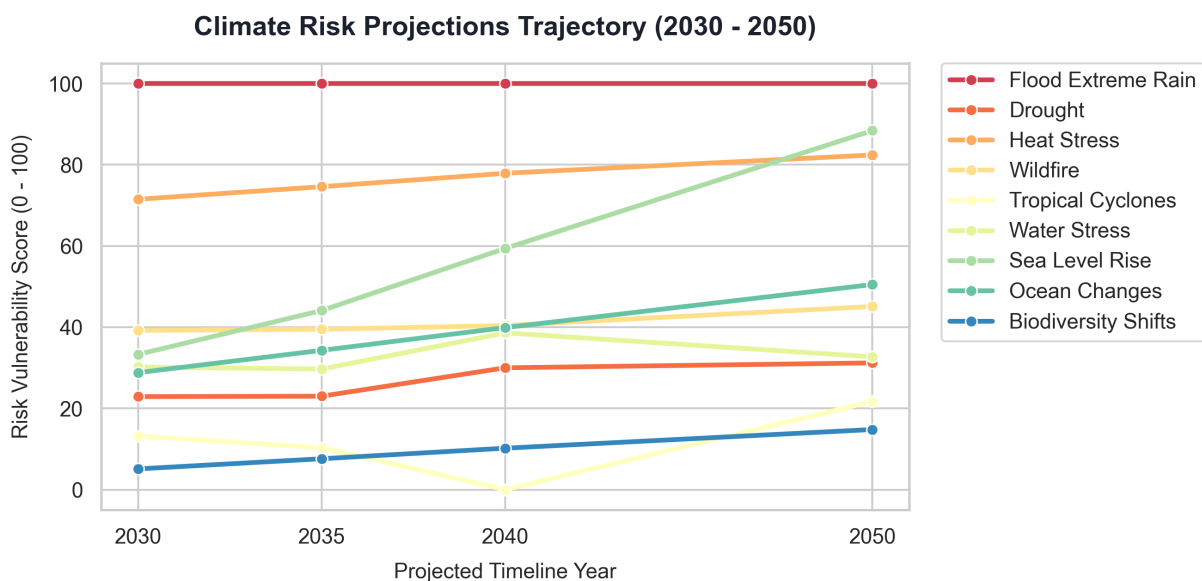
Executive Summary

Asset Phoenix Data Center II (a Data Center) situated at 4100, North 44th Street, Camelback East, Phoenix, Maricopa County, Arizona, 85018, United States has been evaluated under the climate scenario SSP585. Projections up to 2050 indicate vulnerabilities driven primarily by Flood Extreme Rain which registers a high risk score of 100.0 by mid-century. Operational contingency plans should prioritize climate resilience retrofits.

Climate Risk Projections Matrix

Risk Parameter	2030	2035	2040	2050
Flood Extreme Rain	100.0 (High)	100.0 (High)	100.0 (High)	100.0 (High)
Drought	22.9 (Low)	23.0 (Low)	30.0 (Low)	31.2 (Low)
Heat Stress	71.5 (High)	74.6 (High)	77.9 (High)	82.4 (High)
Wildfire	39.2 (Medium)	39.5 (Medium)	40.4 (Medium)	45.1 (Medium)
Tropical Cyclones	13.2 (Low)	10.3 (Low)	0.0 (Low)	21.7 (Low)
Water Stress	30.2 (Low)	29.7 (Low)	38.6 (Medium)	32.7 (Low)
Sea Level Rise	33.3 (Low)	44.1 (Medium)	59.4 (Medium)	88.4 (High)
Ocean Changes	28.8 (Low)	34.3 (Low)	39.9 (Medium)	50.5 (Medium)
Biodiversity Shifts	5.1 (Low)	7.6 (Low)	10.2 (Low)	14.8 (Low)

Climate Projections Trend Chart



Detailed Risk Driver Explanations

Flood Extreme Rain (2050 Status: **High**) — Extreme precipitation trends indicate an increase in short-duration storm frequency. Flooding risk is elevated, threatening storm drainage capacities and low-level storage bays.

Drought (2050 Status: **Low**) — Decreasing seasonal rainfall averages and elevated evapotranspiration suggest more persistent drought patterns. This could strain raw water reservoirs and escalate regional fire warnings.

Heat Stress (2050 Status: **High**) — Projected temperature increases show rising wet-bulb thresholds. Cooling energy demands are expected to spike, while operational capacity could be throttled during peak summer occurrences.

Wildfire (2050 Status: **Medium**) — Climatic fire weather indices indicate extended wildfire seasons. Assets in proximity to dry biomes or industrial zones face higher risk from smoke contamination and convective fire gusts.

Tropical Cyclones (2050 Status: **Low**) — While cyclonic paths remain historically concentrated, modeling predicts a minor expansion of peak wind velocities, posing potential structural risks to cladding and roofing systems.

Water Stress (2050 Status: **Low**) — Baseline water withdrawal vs renewable availability trends show increasing water scarcity. Water-dependent operations (e.g. data center liquid cooling) must explore closed-loop recycling.

Sea Level Rise (2050 Status: **High**) — Coastal baseline levels show progressive high-tide encroachments. Assets under low elevation zones may experience increased localized tidal flooding and saltwater intrusion.

Ocean Changes (2050 Status: **Medium**) — Ocean pH levels show structural acidification trends. Sea surface temperatures (SST) are projected to rise, affecting aquaculture yields and accelerating localized coral bleaching.

Biodiversity Shifts (2050 Status: **Low**) — Local biomes indicate moderate transitions in species richness. Ecosystem services, such as soil stability and local hydrological filtration, may experience localized decline by 2050.

ESG Integration & Target Alignment

Strategic Alignment: Based on the corporate sustainability report, there are partial strategies aligning with carbon footprint reductions. However, operational physical risk adaptations remain generalized.

Identified Mitigation Gaps:

- Lack of local site adaptation pathways for extreme weather scenarios.

ESG Strategic Recommendations:

- Draft site-specific water recycling systems to buffer water stress.
- Reinforce physical storage facilities to withstand storm wind parameters.
- Review carbon offsetting portfolio on Prana Earth Marketplace to offset scope emissions.

Marketplace Adaptation Opportunities

The following adaptation and resilience project recommendations have been generated based on the highest environmental risk categories identified for this asset. Click individual links to explore mitigation details directly on the Prana Earth Marketplace.

Project Title	Category	Description	Marketplace Link
Coastal & Inland Wetlands Restoration	Nature-Based Solution	Establish natural buffer zones to absorb surge runoff, reducing flooding risks by up to 35%.	Explore Project
Industrial Stormwater Management Systems	Engineering	High-capacity drainage filters and detention ponds to protect asset assets from flash precipitation.	Explore Project
Afforestation Carbon Offsetting Scheme	Carbon Mitigation	Standardized carbon credits to offset residual emissions and align with net-zero roadmaps.	Explore Project